

Complex Analysis General Exam, August 2020

General guidelines: In order to receive full credit, a detailed argument (rather than a sketch of the proof) is needed. Whenever applying one of the standard theorems, please indicate that clearly.

Pledge:

Problem 1

Let f be an entire function. Assume that

$$\max_{|z|=r} |f(z)| \leq 10 \log r$$

for $r \geq 100$. Show that f is constant.

Problem 2

Let f be holomorphic on $\Omega = \{0 < |z| < 1\}$, and suppose that

- (i) $|f(z)| > 1$ for all $z \in \Omega$,
- (ii) there exists a sequence $\{z_k\}_{k=1}^{\infty} \subset \Omega$, $z_k \rightarrow 0$, $k \rightarrow \infty$ such that $|f(z_k)| \leq 10$ for $k \geq 100$.

Is 0 a removable singularity?

Problem 3

Compute

$$\int_0^{\infty} \frac{\log x}{x^2 + 1} dx$$

via complex integration. Show all estimates.

Problem 4

Let $f : D = \{|z| < 1\} \rightarrow \mathbb{C}$ be holomorphic. Assume that $f(0) = 0$ and $|\operatorname{Re} f(z)| < 1$, for all $z \in D$. Show that

$$|f'(0)| \leq \frac{4}{\pi}$$

Hint: A function $g(w) = \frac{e^{i\pi\frac{w}{2}} - 1}{e^{i\pi\frac{w}{2}} + 1}$ mapping $\{|\operatorname{Re} w| < 1\}$ conformally to D might prove useful.

Problem 5

Let $\mathcal{F}$ be a family of entire functions with the property that for any circle C , there exists a constant $M_C > 0$ such that

$$\sup_{f \in \mathcal{F}} \max_{z \in C} |f(z)| \leq M_C$$

- (i) show that there exists a sequence $\{f_n\}_{n=1}^{\infty} \subset \mathcal{F}$ converging to an entire function g uniformly on compacts in $\mathbb{C}$
- (ii) suppose that $f_n(z) \neq 7$ for $n \geq 77$ and all $z \in \{|z| < 1\}$. Can $g\left(\frac{1+i}{2}\right) = 7$?